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Extending QFD for smart product-service systems with smartness parameter categorization

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ABSTRACT: A key aspect of Circular Economy (CE) is focusing on value creation through customer functionality and service across the entire product life cycle, supported by digitalization tools for improved management. This shift has led to the rise of smart Product-Service Systems (PSS) models. However, designing smart PSS is complex, requiring methodological support for successful implementation. This study explored the feasibility of a novel tool based on the Quality Function Deployment (QFD) framework through its practical application in the photovoltaic industry.

KEYWORDS: quality function deployment (QFD), circular economy, smart product-service systems, photovoltaic system, smartness parameters

1. Introduction

The growing global focus on circular economy (CE) based business models is fostering the shift toward value creation centered on functionality and service provision throughout the entire product life cycle (Golinska-Dawson et al., 2024; Fargnoli et al., 2012; Bocken et al., 2016). This evolution has given rise to servitization-oriented business models, particularly Product-as-a-Service (PaaS) systems that focus on providing access to product functionality rather than ownership (da Costa Fernandes et al., 2020; Sakao & Nordholm, 2021). Developing effective PaaS solutions demands a holistic and systemic approach that integrates product, service, and supply chain design, ultimately reshaping traditional corporate business models into more sustainable and service-oriented frameworks (Neramballi et al., 2024). The implementation of a PaaS approach offers significant potential, especially for industrial complex systems (e.g., high-tech products) where after-sales services are essential for ensuring product functionality and preventing obsolescence (Bocken et al., 2018). These systems typically involve multiple stakeholders, including third-party service providers, and operate in regulated markets that require compliance with safety and environmental directives throughout the contract duration (Diaz et al., 2021). To succeed, manufacturers/providers must focus on delivering customer value through an optimized mix of goods and service activities (Neramballi et al., 2025). However, the achievement of such a goal is not easy for companies due to numerous barriers to PaaS adoption (Matschewsky et al., 2018). These challenges are particularly pronounced in the electrical and electronic equipment (EEE) sector serving B2C markets, where empirical implementation studies are scarce and the transition to service-based models is underexplored (Golinska-Dawson et al., 2025). Indeed, the operationalization of PaaS practices remains difficult to implement effectively (Pollard et al., 2021).

Recent advancements in Industry 4.0 technologies have demonstrated that some improvements can be achieved in implementing PSS business models through the integration of digital tools, which allow manufacturers/providers to enhance product lifecycle management. These solutions, named Smart Product-Service Systems (Smart-PSS), refer to the integration of smart products and e-services into cohesive, IT-driven solutions, which allow companies to collect real-time data on product usage and customer behaviour, supporting predictive maintenance, personalized service delivery, and product improvement throughout the lifecycle. These offerings leverage advanced information technologies to

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transform data into actionable knowledge and embed services within products to increase responsiveness to customer demands. As observed by Chen et al. (2024), the combined use of advanced digital technologies (e.g., smart sensing, Internet of Things (IoT), big data analytics, digital twin, and artificial intelligence) allows manufacturers to address both digitalization and servitization in their business offerings, enhancing competitiveness, generating business value, and supporting environmental sustainability.

These Information and Communications Technology (ICT) tools enable new functionalities and facilitate the delivery of digital services via physical goods, thereby extending the interaction between manufacturers/providers and customers into the so-called Cyber Physical Systems (CPS) (Machchhar et al., 2022). However, these advances also pose new challenges in data management and decision-making, calling for robust support tools to help engineers manage Smart-PSS complexity. Moreover, to achieve such a goal, the development of systems characterized by numerous heterogeneous and interactive components is needed, requiring structured methodologies that guide their design and evaluation processes (Paliyenko et al., 2024). This represents one of the main barriers in implementing Smart-PSS solutions, i.e. the difficulties in integrating product and service development processes, and those related to embedding smart components into existing and new systems (Le Dain et al., 2023).

As pointed out by Zeng et al. (2025), although an increasing attention to Smart-PSS development can be observed, more research is still needed encompassing case studies and content analyses. More specifically, research gaps remain in the conceptual development stage of Smart-PSS, particularly regarding the comprehensive identification of customer needs and expectations (Liu et al., 2019).

To address these gaps, this study aims to augment knowledge on tools for Smart-PSS development by investigating the use of a QFD-based approach. Actually, while the application of QFD-based approaches in the development of PSS solutions is well-established in the literature (Bertoni, 2024), few studies have addressed the design and development of the smart characteristics inherent in Smart PSS. Among them, we selected the Quality Function Deployment for Smart Product Service Systems (QFDforSmart-PSS) method (Fargnoli & Haber, 2023), which extends a solid QFD framework for PSS design to the development of smart solutions. Built upon the QFD for PSS (QFDforPSS) framework (Arai and Shimomura, 2004), QFDforSmart-PSS integrates Information Technology (IT) driven features alongside PSS characteristics and components. This augmentation allows engineers to better capture, analyze, and prioritize customer requirements, offering a more detailed understanding of their interrelations in Smart-PSS design. However, although its first application showed some positive results, this tool is at its initial stage of development and further research is needed to verify its viability.

With this goal in mind, the current study applied the method to a case study focused on designing a Product-Service System (PSS) business model for urban photovoltaic (PV) systems in collaboration with a company operating in the renewable energy sector. A sensitivity analysis was also conducted to verify the stability and robustness of the results. Accordingly, this paper aims to discuss the advantages of addressing the IT aspect as a separate category from the product or service aspect in smart PSS design.

The remainder of the article is organized as follows: in the next section, the research approach is described. Section 3 illustrates the practical application of the QFDforSmart-PSS method, whose results are discussed in Section 4. Concluding remarks and future research are addressed in Section 5.

2. Materials and methods

The Quality Function Deployment for Smart Product Service Systems (QFDforSmart-PSS) method (Fargnoli & Haber, 2023) is grounded on the Quality Function Deployment for Product Service System (QFDforPSS) approach, which was introduced to enable the simultaneous deployment of product and service features (Sakao & Shimomura, 2007). QFDforPSS comprises two distinct phases, each characterized by a specific House of Quality (HoQ), as shown in Figure 1: the first involves translating customer requirements into Receiver State Parameters (RSPs) to ensure comparability, which are then mapped to PSS characteristics; the second phase links these characteristics to specific PSS components (Sakao et al., 2009).

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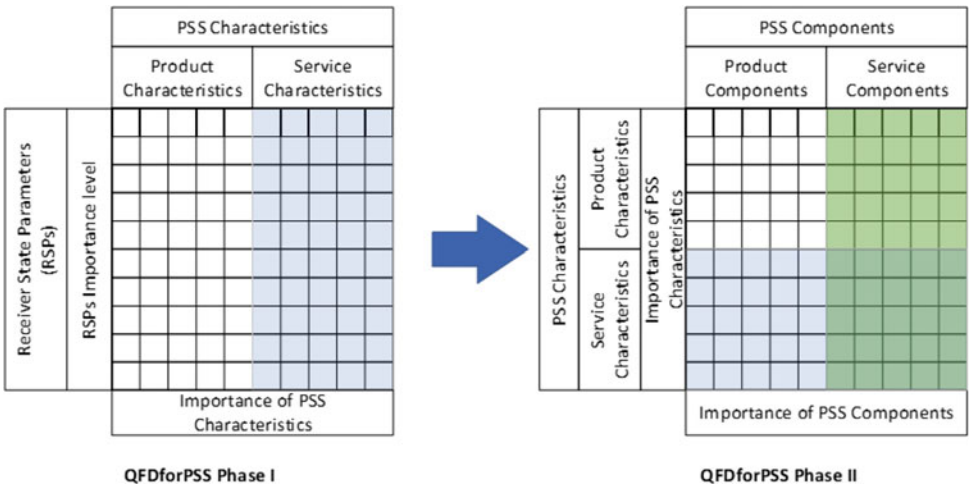


Figure 1. Scheme of the QFDforPSS method (adapted from Fargnoli & Sakao (2017))

This paper investigates a design method, QFDforSmart-PSS, where the first phase of QFDforPSS is augmented through the introduction of the Information Technology Characteristics (ITChs), while in the second phase the Information Technology Components (ITCos) are introduced (Figure 2).

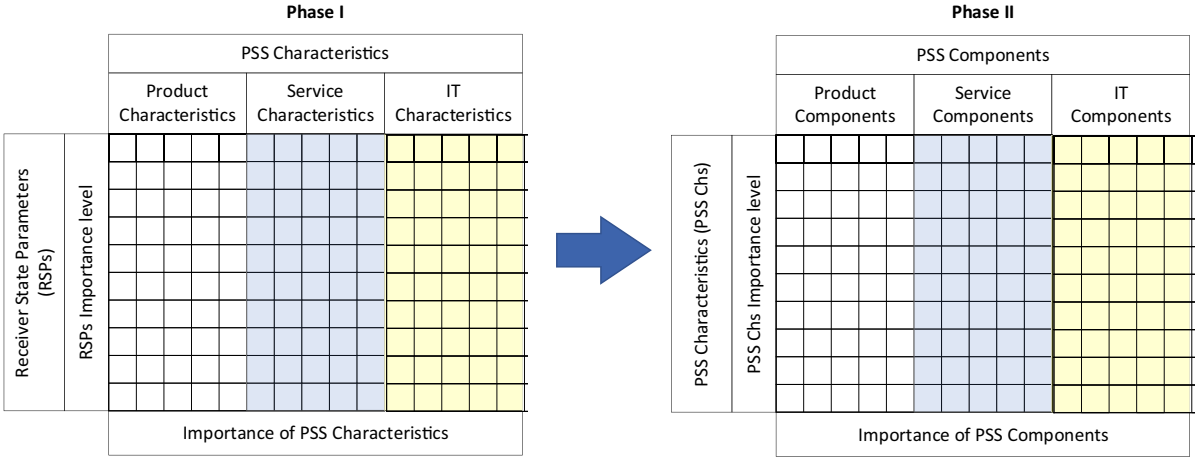


Figure 2. Scheme of the QFDforSmart-PSS method (adapted from Fargnoli & Haber (2023))

The meaning of the main parameters can be explained as follows:

- Receiver State Parameters (RSPs): represent the desired outcomes or expectations of customers and can be classified as either value-oriented or cost-oriented categories (Sakao et al., 2009).
- PSS Characteristics (PSS-Chs): represent the engineer's vision of RSPs, encompassing product (PChs), service (SChs), and information technology (ITChs) characteristics.
- PSS Components (PSS-COs): represent tangible (product components (PCOs)), intangible (service components (PCOs)), and smart (information technology components (ITCos)) elements that constitute a Smart-PSS.

As far as the method's functioning is concerned, the conventional HoQ scoring system (Fargnoli & Sakao, 2017) is adopted, where: a 1-5 scale is used to evaluate the importance level of RSPS; a 0-1-3-9 scale is used to evaluate the strength of the relationships between "hows" and "whats" in each HoQ. The focus of this paper lies in the introduction of the IT elements as the third category, complementing the product and service categories. The IT components are essentially software and software-related enablers (e.g. a cloud storage system or a system integration middleware), whereas the product components can be defined by their dimensional, aesthetic, technological, and functional properties, and service components refer to the resources required for service delivery, categorized into human resources, information, and service tools (Sakao et al., 2017).
Regarding existing PSS design theories, e.g. (Sakao et al., 2017), how to address software is ambiguous. Some literature informs that products are physical goods (e.g., furniture and vehicles) and services are

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human activities (e.g., repair and consultation) (Sakao et al., 2022). On the one hand, software as a design outcome is not a human activity, which has inseparability of production and use. On the other hand, it can be designed, produced, and then used even after the product has entered its use phase, as exemplified by the widely recognized concept of software-defined vehicles. It makes the PSS solution smart(er) and plays a critical role in enabling intelligent capabilities, although its absence does not affect the basic functions. Software has a distinct feature from a product and a service. Hence, PSS design theories will benefit from treating software as neither a product nor a service by using and advancing existing knowledge without inconsistency. Moreover, previous studies (Fargnoli and Haber, 2023; Liu et al., 2019) identify a gap in the systematic identification and prioritization of smart attributes in Smart-PSS design, as digital capabilities cut across traditional product–service boundaries and are often underrepresented when embedded within them. Capturing IT elements as a separate category addresses this limitation by enabling a clearer and more granular assessment of smart functionalities and is supported by case study evidence showing their central role in customer value creation. Based on this reasoning, this paper proposes the IT elements as a separate category from products and services in PSS design.

3. Case study

The method was applied to the design of a novel PSS business model in the field of PV systems for urban and suburban contexts. Recent research (Simionescu & Radulescu, 2025) has brought to light that there is a growing demand for PV systems in urban areas due to multiple factors such as: increasing environmental concerns, increasing energy costs, and governmental subsidies for the adoption of renewable energy sources. The analysis of this emerging market has also outlined not only the importance of technical specifications but also the growing relevance of customer-centric factors such as the aesthetic compatibility with the building, as well as the ease of use and integration with existing energy infrastructure. Moreover, the provision of services for the management of the whole life cycle of the PV system has emerged as an important business opportunity for providers (IEA, 2024).

The practical application of the QFDforPSS method was performed in collaboration with a company operating as an energy provider offering PV solutions to private users. The current business model encompasses installation and corrective maintenance interventions within the warranty period, while other services such as preliminary design of the system integration, preventive maintenance and end-of-life take-back should be purchased separately when acquiring the PV system. The company is willing to expand its business by including more integrated product-service solutions, such as PaaS models. The company is motivated by the increasing number of requests from customers, thanks to the tax relief available for installing PV systems on residential properties. As far as data collection is concerned, it must be noted that customer needs were retrieved from previous research (Fargnoli et al., 2024) and further elaborated through consultations with a group of four experts from the company belonging to both the customer care and project management units. A final meeting with the company experts allowed us to discuss the results achieved.

3.1. Phase I

Starting from the analysis of data collected, the most relevant customer expectations were taken into account and then translated into the following list of RSPs. Besides, a list of PSS Characteristics was defined based on the criteria exposed in the previous section. Regarding the scoring approach used, individual consultations with experts led to four different importance ratings, whose mean values were used in the first HoQ of the method. Differently, for the weighting of the strength between RSPs and PSS Characteristics, scores were obtained as a result of a common discussion with experts. The final output of Phase I is illustrated in Figure 3.

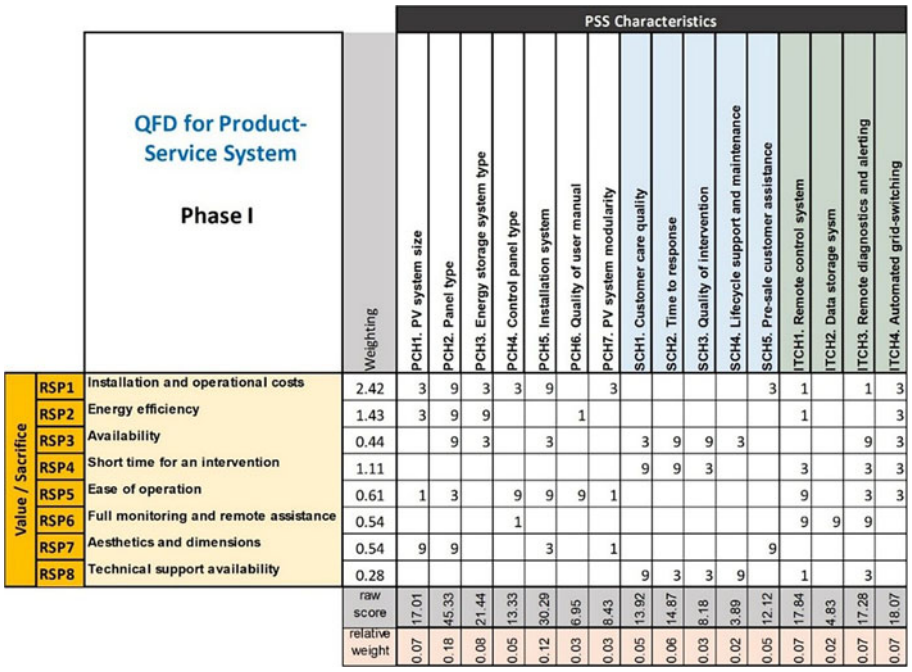
3.2. Phase II

The second phase of the method concerned the translation of PSS Characteristics into components and their prioritization. The HoQ scoring was carried out as in Phase I, and the prioritization results are shown in Figure 4, where the relative importance of PSS Components is outlined.

3.3. Sensitivity analysis

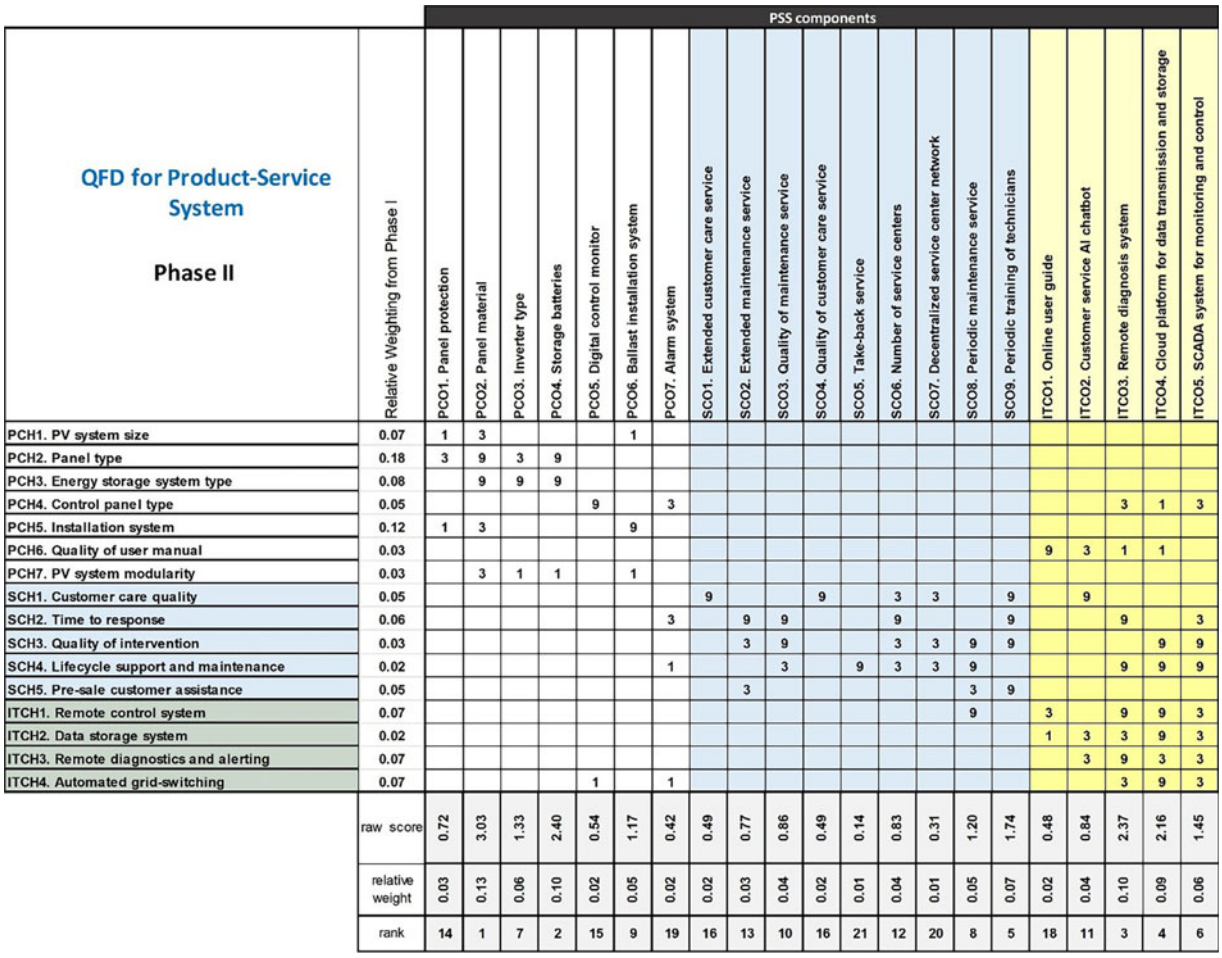
To verify the robustness of the proposed approach, a sensitivity analysis was carried out using the criteria proposed by Wang et al. (2020).

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QFD for Product-Service System — Phase I			Weighting	PCH1. PV system size	PCH2. Panel type	PCH3. Energy storage system type	PCH4. Control panel type	PCH5. Installation system	PCH6. Quality of user manual	PCH7. PV system modularity	SCH1. Customer care quality	SCH2. Time to response	SCH3. Quality of intervention	SCH4. Lifecycle support and maintenance	SCH5. Pre-sale customer assistance	ITCH1. Remote control system	ITCH2. Data storage sysm	ITCH3. Remote diagnostics and alerting	ITCH4. Automated grid-switching
				PSS Characteristics															
Value / Sacrifice	RSP1	Installation and operational costs	2.42	3	9	3	3	9		3					3	1		1	3
	RSP2	Energy efficiency	1.43	3	9	9			1							1			3
	RSP3	Availability	0.44		9	3		3			3	9	9	3				9	3
	RSP4	Short time for an intervention	1.11								9	9	3			3		3	3
	RSP5	Ease of operation	0.61	1	3		9	9	9	1						9		3	3
	RSP6	Full monitoring and remote assistance	0.54				1									9	9	9	
	RSP7	Aesthetics and dimensions	0.54	9	9			3		1					9				
	RSP8	Technical support availability	0.28								9	3	3	9		1		3	
			raw score	17.01	45.33	21.44	13.33	30.29	6.95	8.43	13.92	14.87	8.18	3.89	12.12	17.84	4.83	17.28	18.07
			relative weight	0.07	0.18	0.08	0.05	0.12	0.03	0.03	0.05	0.06	0.03	0.02	0.05	0.07	0.02	0.07	0.07

Figure 3. Phase I of QFDforSmart-PSS method



QFD for Product-Service System — Phase II	Relative Weighting from Phase I	PCO1. Panel protection	PCO2. Panel material	PCO3. Inverter type	PCO4. Storage batteries	PCO5. Digital control monitor	PCO6. Ballast installation system	PCO7. Alarm system	SCO1. Extended customer care service	SCO2. Extended maintenance service	SCO3. Quality of maintenance service	SCO4. Quality of customer care service	SCO5. Take-back service	SCO6. Number of service centers	SCO7. Decentralized service center network	SCO8. Periodic maintenance service	SCO9. Periodic training of technicians	ITCO1. Online user guide	ITCO2. Customer service AI chatbot	ITCO3. Remote diagnosis system	ITCO4. Cloud platform for data transmission and storage	ITCO5. SCADA system for monitoring and control
		PSS components																				
PCH1. PV system size	0.07	1	3				1															
PCH2. Panel type	0.18	3	9	3	9																	
PCH3. Energy storage system type	0.08		9	9	9																	
PCH4. Control panel type	0.05					9		3												3	1	3
PCH5. Installation system	0.12	1	3				9															
PCH6. Quality of user manual	0.03																	9	3	1	1	
PCH7. PV system modularity	0.03		3	1	1		1															
SCH1. Customer care quality	0.05								9			9		3	3		9		9			
SCH2. Time to response	0.06							3		9	9			9			9			9		3
SCH3. Quality of intervention	0.03									3	9			3	3	9	9				9	9
SCH4. Lifecycle support and maintenance	0.02							1			3		9	3	3	9				9	9	9
SCH5. Pre-sale customer assistance	0.05									3						3	9					
ITCH1. Remote control system	0.07															9		3		9	9	3
ITCH2. Data storage system	0.02																	1	3	3	9	3
ITCH3. Remote diagnostics and alerting	0.07																		3	9	3	3
ITCH4. Automated grid-switching	0.07					1		1												3	9	3
	raw score	0.72	3.03	1.33	2.40	0.54	1.17	0.42	0.49	0.77	0.86	0.49	0.14	0.83	0.31	1.20	1.74	0.48	0.84	2.37	2.16	1.45
	relative weight	0.03	0.13	0.06	0.10	0.02	0.05	0.02	0.02	0.03	0.04	0.02	0.01	0.04	0.01	0.05	0.07	0.02	0.04	0.10	0.09	0.06
	rank	14	1	7	2	15	9	19	16	13	10	16	21	12	20	8	5	18	11	3	4	6

Figure 4. Phase II of QFDforSmart-PSS method

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The core of the analysis involves the construction of a set of discrete weighting scenarios for each HoQ. For Phase I, 8 different scenarios were developed corresponding to the RSPs and within each scenario, a single RSP was subjected to an increase in importance weight (+35%), while the weights of all remaining criteria were commensurately reduced (-35%). For example, the second scenario S2 concerned an increase in the importance of RSP2, while the values of the other RSPs were reduced. In Figure 5, the result of this analysis is shown, where S0 refers to the original prioritization of PSS Characteristics (i.e. the baseline).

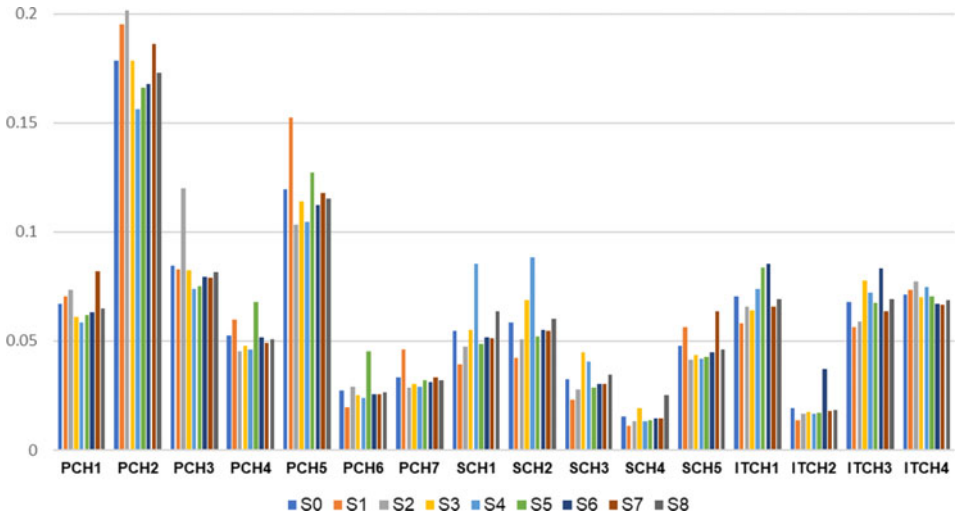


Figure 5. Comparison of the importance of PSS Characteristics in different scenarios

Then, to quantitatively assess the stability of the alternative rankings across the 8 weighting scenarios, Spearman's Rank Correlation Coefficients (SCC) were computed. The results of these correlations are systematically presented in Table 1.

Table 1. Spearman's rank correlation coefficients for sensitivity analysis in phase I

	S0	S1	S2	S3	S4	S5	S6	S7	S8
S0	1	0.97	0.97	0.99	0.94	0.98	0.98	0.99	1
S1		1	0.93	0.94	0.86	0.96	0.94	0.97	0.96
S2			1	0.95	0.88	0.93	0.94	0.96	0.96
S3				1	0.95	0.95	0.97	0.97	0.99
S4					1	0.90	0.92	0.91	0.95
S5						1	0.97	0.95	0.97
S6							1	0.96	0.97
S7								1	0.98
S8									1

The analysis reveals a high degree of rank correlation between the baseline ranking and those generated under the perturbed weight scenarios. Although the intentional modification of criteria weights resulted in minor adjustments to the final rank of some characteristics, these changes were determined to be statistically insignificant in terms of overall concordance, as the majority of SCCs exceed 0.95, indicating a very strong positive association, and the minimum observed SCC across all scenarios is greater than 0.86, confirming a substantial correlation even in the most perturbed scenario.

For Phase II, the different weights of PSS Characteristics in the scenarios from Phase I were used to develop eight different HoQs and thus 8 different rankings of PSS Components, whose comparison is shown in Figure 6. Spearman's Rank Correlation Coefficients were therefore computed to verify the robustness of the results (Table 2). The output of this analysis showed strong correlations between the various scenarios.

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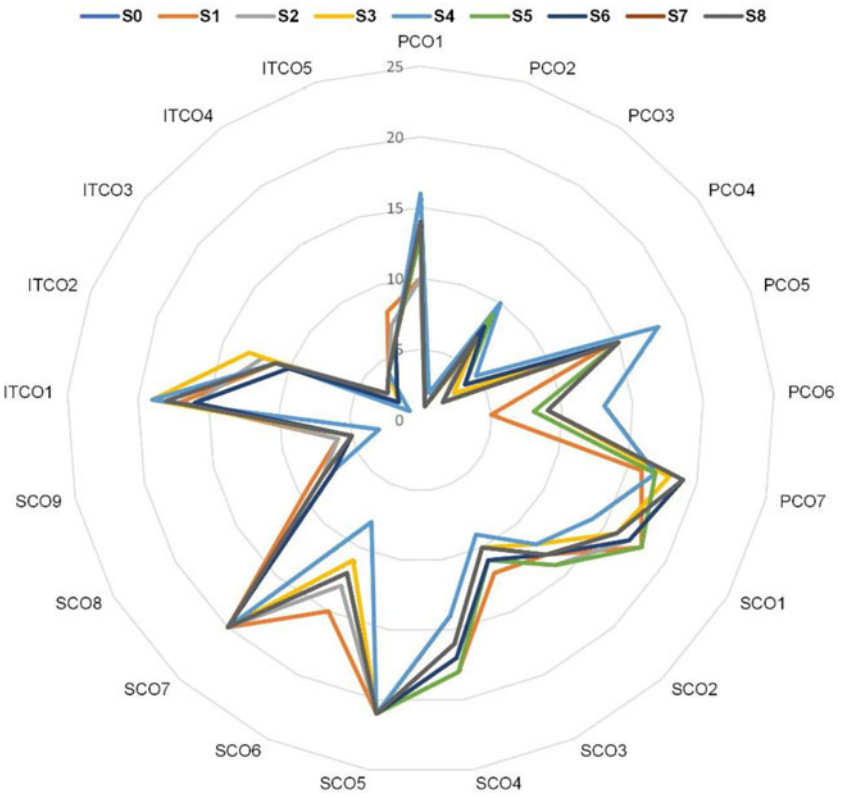


Figure 6. Comparison of the importance of PSS components in different scenarios

Table 2. Spearman's rank correlation coefficients for sensitivity analysis in phase II

	S0	S1	S2	S3	S4	S5	S6	S7	S8
S0	1	0.95	0.98	0.99	0.94	0.98	0.99	1	1
S1		1	0.97	0.95	0.83	0.97	0.95	0.95	0.95
S2			1	0.97	0.88	0.97	0.97	0.98	0.98
S3				1	0.95	0.97	0.98	0.99	0.99
S4					1	0.92	0.94	0.94	0.94
S5						1	0.99	0.98	0.98
S6							1	0.99	0.99
S7								1	1
S8									1

4. Discussion

The implementation of the QFDforSmart-PSS method brought to light that the most relevant customer needs and expectations refer to installation and operational costs (RSP1), energy efficiency (RSP2), and short time for intervention (RSP4). Conversely, the system durability (RSP3) and the quality of technical support (RSP8) were considered the least important adding-value features. This can be explained by the fact that RSPs were derived from a survey among customers willing to purchase a PV system, and most companies today offer the system with basic warranties, excluding damages resulting from improper installation or lack of maintenance. The output of the first phase of the method shows the prevalence of product features over the service and IT ones, although among the latter ITCH4, ITCH1, and ITCH3 exhibit a good ranking. At the same time, the importance given to the panel type (PCH2) and installation system (PCH5), which are both a challenge for designers. Actually, the specific combination of high-efficiency system, reduced dimensions, and building integration represents a major concern for the company technicians.

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A similar trend is observed when examining the relevance of PSS components, as features such as ITCO3 (remote diagnosis system), ITCO4 (cloud platform for data transmission and storage), and ITCO5 (Supervisory Control and Data Acquisition (SCADA) system for monitoring and control) were identified as among the most critical elements for implementation. The higher level of granularity of the results of this phase allows us to bring to light relevant issues for engineers operating in this sector. More specifically, considering the product attributes, the materials of panels and those of the storage system (i.e. batteries) represent a criticality that can certainly be related to the potential supply shortages of their components and materials (Calderon et al., 2024). Overall results show that customers value ease of management and long-term sustainability, including maintenance and end-of-life handling. These preferences align with evolving recycling and waste management policies, highlighting the need for PV providers to adopt life-cycle management strategies (IEA, 2024; Saedpanah et al., 2020). From the service perspective, on the one hand, the quality of customer support in the use phase of the system emerges as paramount to implement a PaaS approach, especially in terms of the company's capabilities. On the other hand, aspects such as reverse logistics for the collection of PV systems after their use appear not relevant. This output could be explained by the fact that, together with the recent legislative framework, this type of market is still immature in most EU countries (Radavičius et al., 2025).

The study contributed to verifying the effectiveness of a novel QFD-based method for the development of smart PSS solutions. The more structured conversion of customer requirements guaranteed by QFDforSmartPSS facilitates a comprehensive enhancement of both the physical and service-oriented aspects, fostering the integration of technological elements related to IT, such as cloud computing and remote monitoring, which can be valuable not only in the creation of innovative solutions but also in improving existing systems, i.e. PSS revamping. In terms of design, production, and use as temporal phases, it is technically possible to delay the design of IT elements even to the phase of product use. This advantage has implications on design management: namely, it is enabled to find a set of optimal design activities of products, services, and IT, including when the design is carried out, from a larger solution space. For instance, depending on the available human resources of the designing organization and market needs at a point in time, the IT design can be delayed within a larger time window without delaying the product launch to the same extent. The IT elements that were included in the information category of service components in conventional PSS design methods e.g., (Sakao et al., 2017) can be separately deployed through this novel approach. Thanks to this, this new approach is capable to show various elements of PSS design more clearly to a design team. An example from the presented case is that the remote diagnosis system (ITCO3), belonging to the IT category, can be designed and launched after the product has begun operation without this IT element. Thus, the method can support the evolution of PSS systems toward "smart" configurations that enhance value for both customers and manufacturers or service providers. The results achieved showed a high level of consistency in rankings, and their stability demonstrated that the method can provide credible and reliable outputs. This indicates that the method produces stable and robust prioritization results even under significant weight perturbations. On the other hand, the observed stability may indicate limited responsiveness due to strong interdependencies among product, service, and IT elements.

From a methodological perspective, while existing Smart-PSS design methods, including QFD-based ones, contribute valuable perspectives, such as performance evaluation, innovation support, or service process modelling, they typically treat digital technologies as embedded enablers or evaluation criteria rather than as key design elements. Differently, the proposed approach supports clearer trade-off analysis between physical, service, and IT elements, and holistically addresses the increased complexity of Smart-PSS design. The method can be adapted across industries by tailoring its smart, product, and service elements to sector-specific conditions, as was shown in this study. The method is also expected to scale to different levels of digital maturity and support both new smart solutions and retrofitting of existing systems. This flexibility enhances its applicability, allowing diverse sectors to better integrate smart features and align solutions with customer expectations and industry constraints.

Besides these positive outcomes, some limitations must also be outlined. First, the method suffers from the limitations of the HoQ metrics, which are common to all QFD-based approaches (Fargnoli & Sakao, 2017). Hence, further refinements are needed through the use of supporting multicriteria decision-making tools such as TOPSIS, AHP, and ANP, or through techniques aimed at reducing human judgment bias, such as fuzzy logic. Furthermore, the generalizability of the research findings is constrained by the study's reliance on a single case study and input from only four internal experts, which is a limitation for the generalizability of the results, and the work is therefore framed as exploratory.

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Thus, to strengthen the external validity of the study, future research should incorporate multiple case studies across different sectors, enabling a more comprehensive understanding of Smart-PSS design principles and their relevance in different environments.

5. Conclusion

The development of Smart-PSS models is emerging as a means to foster value creation through customer functionality and service across the entire product life cycle. This study examined a novel approach to support designers, which expands the features of QFD to develop and compare product, service, and information technology elements. Thus, despite the exploratory nature of this research, it contributes to augmenting knowledge on QFD-based methods for PSS development, providing practical insights for their further evolution.

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